12.21 Breaking a Stick

WORKSHEET

The question

(i) A rod of unit length is cut into pieces of length X and $1-X$; the latter is then cut in half. The random variable X is uniformly distributed over $[0,1]$. For some values of X a triangle can be formed from the three pieces of the rod. Show that the conditional probability that, if a triangle can be formed, it will be obtuse-angled is $3-2\sqrt{2}$.

(ii) The bivariate distribution of the random variables X and Y is uniform over the triangle with vertices $(1,0)$, $(1,1)$ and $(0,1)$. A pair of values x, y is chosen at random from this distribution and a (perhaps degenerate) triangle ABC is constructed with $BC = x$ and $CA = y$ and $AB = 2-x-y$. Show that the construction is always possible and that $\angle ABC$ is obtuse if and only if

$$y > \frac{x^2 - 2x + 2}{2 - x}.$$

Deduce that the probability that $\angle ABC$ is obtuse is $3 - 4\ln 2$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Three lengths form a triangle if and only if each is less than the sum of the other two. If the lengths add up to L, what does this say about the longest one? When is the angle opposite side c obtuse?

STAGE 2 The second construction

Ask yourself Check the three triangle inequalities for sides x, y and $2-x-y$, for (x,y) in the triangle with vertices $(1,0)$, $(1,1)$, $(0,1)$.

Ask yourself The angle at B is opposite the side $CA = y$. Write the condition $y^2 > x^2 + (2-x-y)^2$ as an inequality for y.

STAGE 3 The area

Ask yourself Divide $x^2 - 2x + 2$ by $2 - x$. Then find the area of the part of the region above the curve, and divide by the region's area, $\frac{1}{2}$.

Try this next

1. **The classic.** Show that if a stick is broken at two independent uniform points, the probability that the pieces form a triangle is $\frac{1}{4}$.

2. **Break the longer piece.** Break a stick at a uniform point, then break the *longer* piece at a uniform point. Show that the probability that the three pieces form a triangle is $2\ln 2 - 1 \approx 0.386$.

3. **Acute or obtuse.** In part (ii), find the probability that the triangle ABC has an obtuse angle at C (opposite $AB = 2-x-y$). Hence find the probability that it has an obtuse angle somewhere.

4. **Expected longest piece.** For the classic broken stick, show that the expected length of the longest piece is $\frac{11}{18}$.

Further reading

- **Martin Gardner, *The Colossal Book of Mathematics*** (Norton), which includes the broken-stick problem.

- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover).
- **Lewis Carroll, *Pillow Problems*** (Dover, with *A Tangled Tale*). Victorian puzzles, including random triangles.

The full guided solution, and what the question leads to at university, is in Section 12.21:
https://beyondstep.keithmcnulty.org/ch12/broken-stick.html